

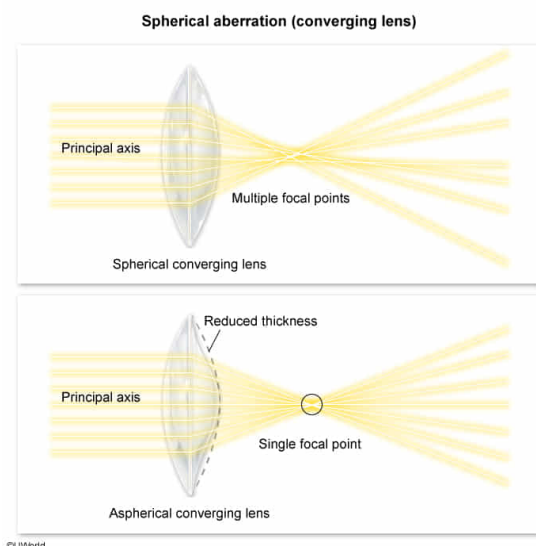
The spherical aberration of a converging lens can be corrected by reducing the thickness of the lens periphery. The spherical aberration of the lens is caused because light rays emerging from the lens:

- ☐ A. are insufficiently refracted at the lens periphery. [21%]
- ☐ B. converge at the lens focal point uniformly. [11%]
- ✓ ☒ C. are excessively refracted at the lens periphery. [54%]
- ☐ D. are scattered due to chromatic dispersion. [12%]

Correct

54% answered correctly

Explanation:



An **ideal lens** generates an image of an object at a single focal point, regardless of the locations at which the light rays enter and exit the lens. **Spherical aberration** describes the phenomenon by which real lenses' perfectly rounded (ie, spherical) surfaces do not produce an image at a single point, but rather at a series of **focal points**.

In converging lenses, the location of each focal point is related to the distance from the **principal axis** (ie, the lens center) at which light enters and exits the lens. Spherical aberration is most pronounced among rays entering and exiting the **lens periphery** (ie, distant from the principal axis). Furthermore, correcting spherical aberration in converging lenses requires using an aspherical lens in which the thickness of the lens periphery is *decreased* relative to a perfectly rounded lens.

Because reducing the thickness of the lens periphery will lead to *less* refraction of light, light rays exiting the periphery of a spherical converging lens can be said to refract excessively, converging on a focal point that is too close to the lens.

(Choice A) Spherical aberration in a converging lens is characterized by refractive events that focus light emerging at the lens periphery more dramatically than light emerging closer to the lens center. Therefore, light is *excessively*, *not* insufficiently, refracted at the lens periphery.

(Choice B) All lens and mirror aberrations describe a *failure* of optical instruments to converge or diverge light rays uniformly.

(Choice D) **Chromatic dispersion** describes the failure of a lens to focus multicolored light onto a single point. Chromatic dispersion in converging lenses is **corrected** by using a diverging lens to *increase* the thickness of the lens periphery.

Educational objective:

Spherical aberration occurs when lenses with perfectly rounded surfaces focus light at multiple focal points. Spherical aberration is most pronounced among light rays entering and exiting the periphery of converging lenses.

Time spent:0 sec

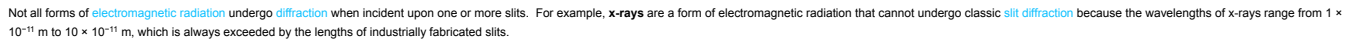
Subject:
Physics

QID:402484

System:
Light and Sound

✔ ☒ A. determination of three-dimensional molecular structure. [63%]
☐ B. determination of the work function corresponding to each electron. [5%]
☐ C. measuring the wavelength of electromagnetic radiation. [19%]
☐ D. measuring the linear polarization of light. [11%]

Explanation:



When two x-ray waveforms approach and depart a crystalline lattice at an identical incident angle, the waveforms may constructively interfere such that a bright spot is observed on a piece of film distant to the material. The occurrence of constructive interference is a function of the incident angle (θ), the distance between atoms (d), and an integer multiple ($n = \pm 1, \pm 2, \pm 3 \dots$) of the x-ray wavelength (λ):

Because the crystalline lattice of many materials exceeds the complexity of **simple crystals**, the pattern of bright spots produced on the photographic film is indicative of a molecule's **three-dimensional structure** and its **packing orientation** within the crystal lattice.

(Choice C) Using a reference substance with known three-dimensional structure might allow the wavelength of x-ray waveforms to be determined experimentally using x-ray crystallography. However, other diffractive techniques can more easily determine the wavelength of incident light.

Educational objective:

Educational Objective: X-rays diffract within molecules because the space between atoms is comparable to the wavelength of x-rays. X-ray diffraction through a sample of a purified and crystallized material can be used to determine its three-dimensional molecular structure and packing.

QID:402485
System:
Light and Sound

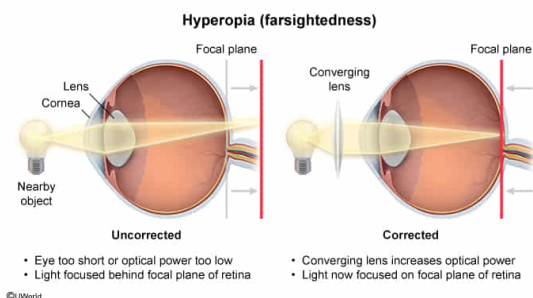
Which of the following will occur if a corrective lens is placed in front of an eye that is unable to form a clear image of a nearby object?

- ☐ A. The focal length will shift away from the eye's lens. [35%]
- ✓ ☒ B. The focal length will shift closer to the eye's lens. [56%]
- ☐ C. Spherical aberration will increase. [4%]
- ☐ D. Optical power will decrease. [3%]

Correct

56% answered correctly

Explanation:



The primary optical components of the eye are the **cornea** and **lens**, both of which form a biological **converging lens** that focuses incoming light rays onto the **retina**, a structure analogous to the film or sensor within a camera. **Refractive errors** occur when the eye is unable to focus incoming light rays onto the retina, producing unclear images.

Refractive errors are caused by a mismatch between the physical dimensions (eg, length) and optical properties of the eye. For example, **hyperopia** (ie, farsightedness) is a condition that results when the **optical power (S)** of the eye is insufficient to refract light rays from **nearby objects**. A hyperopic eye cannot sufficiently refract light rays that approach the eye at an angle (eg, light rays from nearby objects), producing an image of nearby objects that is sharpest at a location *behind* the retina.

Hyperopia is corrected by placing a converging lens in front of the eye, forming an optical system with **greater optical power**. Consequently, placing a corrective lens in front of the hyperopic eye shifts the focal length towards the eye's lens and light from nearby objects is focused onto the retina.

(Choice A) **Myopia** (ie, nearsightedness) describes a visual condition in which the eye forms an image of distant objects at a focal point *in front* of the retina. Correcting myopia (*not* hyperopia) shifts the image away from the lens.

(Choice C) **Spherical aberration** refers to the optical deficiencies of lenses with perfectly spherical surfaces. The surfaces of the cornea or lens are not perfectly spherical, so no spherical aberration occurs.

(Choice D) Placing a converging lens in front of the eye *increases* the optical power of the eye, allowing it to properly focus light rays emitted from or reflected off of nearby objects.

Educational objective:

Hyperopia describes the refractive error that results in an individual being unable to see nearby objects. Hyperopia is caused by the optical power of the eye being insufficient relative to the length of the eye; the condition is corrected by placing a converging lens in front of the eye.

Time spent:0 sec**Subject:**

Physics

QID:402486**System:**

Light and Sound